

Renewable Energy Analysis

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1 Introduction

The transition to more sustainable energy systems is a major global challenge. Renewable energy adoption, reliance on fossil fuels, energy consumption, and electricity-related carbon emissions differ significantly between countries and have changed substantially over time.

The project uses historical energy data [1] to examine these patterns across countries and over time. In a broader context, this type of analysis can provide useful insight into national and regional energy trends, differences in energy consumption, and the progress of the transition toward more sustainable energy systems.

By identifying changes in energy use and comparing countries over time, the analysis aims to improve understanding of how energy systems are evolving and where opportunities may exist for more environmentally sustainable approaches to energy production and consumption.

2 Research Questions and Objectives

The main objective of this project is to develop a reproducible data pipeline for examining global energy-transition trends using the Our World in Data Energy dataset [1].

The analysis addresses the following research questions:

- Which countries have the highest renewable share of electricity?
- Which countries have reduced fossil fuel dependence over time?
- How does energy consumption per person vary between countries?
- Which countries had the highest electricity carbon intensity and how much did they reduce it?

3 Data Source and Dataset Description

The analysis uses the Energy dataset published by Our World in Data (OWID) [1]. The dataset contains historical information about energy production, energy consumption, electricity generation, and carbon emissions for countries and regions around the world.

The current version of the dataset contains 23,377 rows and 130 columns. The observations cover the period from 1900 to 2025, although the available time range differs between countries and variables. Each row represents a country or region for a specific year.

The dataset includes identifying fields such as the country name, country code, and year, as well as measurements related to renewable energy, fossil fuels, electricity generation, population, energy consumption, and carbon intensity.

For this project, only the variables relevant to the research questions will be selected. These include renewable energy consumption and its share of total energy use, fossil-fuel consumption and its share of total energy use, renewable and fossil electricity generation, energy consumption per capita, defined as average consumption per person, population, and electricity carbon intensity.

The dataset contains both individual countries (220) and regional or global aggregates (94). These categories will be examined separately so that comparisons between individual countries are not mixed with aggregated regional or global data.

4 Data Requirements and Variable Selection

To answer the research questions, the project focuses on the variables that describe the composition of energy use, electricity generation, and emissions across countries and over time. The selected variables include country name, country code, year, population, energy consumption per capita, renewable and fossil-fuel consumption, renewable and fossil-fuel electricity generation, and electricity carbon intensity.

These variables were chosen because they allow the analysis to compare how countries differ in their energy mix and how those patterns change over time. Renewable energy indicators make it possible to measure adoption and growth, while fossil-fuel indicators show the extent of dependence on non-renewable sources. Population and energy consumption per capita are important for making fairer comparisons between countries of different sizes. Finally, electricity carbon intensity helps illustrate how emissions related to electricity generation have evolved.

The dataset is not used in its full form. Instead, only the fields relevant to the project questions are retained, which makes the analysis clearer and more focused.

5 Methodology

The raw data were first loaded in Python using pandas. The next step was to remove non-country observations and keep only rows with a valid country code, since the dataset includes both countries and regional or global aggregates. After that, the selected variables were checked for missing values, duplicated entries, and invalid year values.

The cleaned dataset was then structured as a country-year table, where each row represents one country in one year. This structure is suitable for both time-series analysis and cross-country comparisons. Once the data were validated, they were imported into PostgreSQL for SQL-based analysis.

The analytical work was carried out with SQL queries to compare countries, identify years of interest, and measure changes over time. This part of the pipeline is important because it turns the raw dataset into a clearer and more usable analytical dataset.

6 Analysis and Findings

6.1 Which countries have the highest renewable share of electricity?

The first analysis examines which countries have the highest share of electricity generated from renewable sources. This is a useful way to identify countries whose electricity systems are already strongly based on renewable energy.

The 2023 dataset provides the broadest country coverage and is therefore used as the main comparison year. In this year, several countries reported renewable electricity shares of 100%, including Bhutan, Paraguay, Ethiopia, Iceland, and the Central African Republic. Costa Rica and Norway also remained near the top of the ranking, with renewable electricity shares of 99.91% and 98.47%, respectively.

Table 1: Countries with the highest renewable share of electricity in 2023

Country	ISO code	Renewable (%)	Fossil (%)
Bhutan	BTN	100.00	0.00
Paraguay	PRY	100.00	0.00
Ethiopia	ETH	100.00	0.00
Iceland	ISL	100.00	0.00
Central African Republic	CAF	100.00	0.00
Nepal	NPL	100.00	0.00
Albania	ALB	100.00	0.00
Democratic Republic of the Congo	COD	100.00	0.00
Costa Rica	CRI	99.91	0.09
Norway	NOR	98.47	1.53

A smaller but more recent snapshot is also available for 2025, although it covers fewer countries than 2023. In 2025, Costa Rica, Ethiopia, and Paraguay again reached renewable electricity shares of 100%, while Norway, Uruguay, and Denmark remained close to the top of the ranking.

Table 2: Countries with the highest renewable share of electricity in 2025

Country	ISO code	Renewable (%)	Fossil (%)
Costa Rica	CRI	100.00	0.00
Ethiopia	ETH	100.00	0.00
Paraguay	PRY	100.00	0.00
Norway	NOR	98.99	1.00
Uruguay	URY	97.82	2.18
Tajikistan	TJK	94.37	5.63
Luxembourg	LUX	91.56	8.44
Denmark	DNK	91.17	8.83
Kenya	KEN	90.03	9.97
New Zealand	NZL	88.53	11.47

The result shows that some countries already operate electricity systems that are almost entirely renewable according to the dataset. This pattern is visible in several countries that report renewable electricity shares of 100%, although the exact generation mix behind these values is not examined in this project.

This finding is meaningful because it highlights that the transition is already advanced in several countries, while others remain more dependent on fossil-fuel electricity generation. The variation between countries suggests that the transition depends not only on renewable potential, but also on national infrastructure, policy, and the structure of the energy system.

6.2 Which countries have reduced fossil-fuel dependence over time?

The second analysis examines how the fossil-fuel share of total energy use has changed over time for each country. Rather than comparing all countries to a single start and end year, the analysis compares each country’s first available observation with its most recent available observation. This approach accounts for differences in data coverage across countries and provides a more reliable measure of transition over time.

The results show that several countries experienced substantial reductions in fossil-fuel dependence. Iceland, France, Denmark, and Finland recorded the largest declines, with reductions of 56.67, 43.95, 41.68, and 40.38 percentage points, respectively. For the countries in the table, the comparison is between 1965 and 2024, which means the dataset captures a long-term transition rather than only a short-term change.

Table 3: Countries with the largest reduction in fossil-fuel dependence between their first and last available observations

Country	ISO code	Fossil (1965) (%)	Fossil (2024) (%)	Reduction (%)
Iceland	ISL	76.17	19.50	56.67
France	FRA	90.09	46.14	43.95
Denmark	DNK	99.96	58.28	41.68
Finland	FIN	77.15	36.77	40.38
Sweden	SWE	66.46	27.79	38.67
Bulgaria	BGR	95.35	63.30	32.06
Slovakia	SVK	94.20	63.88	30.32
Hungary	HUN	99.89	72.18	27.71
Czechia	CZE	98.68	71.69	26.99
Romania	ROU	98.99	73.31	25.69

These findings suggest that some countries have made noticeable progress in reducing their dependence on fossil fuels. At the same time, the results also show that the transition is uneven across

countries, with some remaining more dependent on fossil sources even in the most recent data.

6.3 How does energy consumption per person vary between countries?

The third analysis compares energy consumption per person across countries. This is important because total national consumption can be misleading when countries have very different population sizes. Using energy consumption per person provides a more comparable measure of how much energy each country consumes on average.

The results show a clear difference between countries. In 2023, Qatar recorded the highest energy consumption per person, followed by Singapore, Iceland, and Bahrain. These values are substantially higher than those of many other countries, suggesting that energy use is concentrated at a much higher level in these economies.

Table 4: Top countries by energy consumption per person in 2023

Country	ISO code	Energy consumption per person
Qatar	QAT	208038.81
Singapore	SGP	171241.08
Iceland	ISL	165312.66
Bahrain	BHR	164932.52
United Arab Emirates	ARE	134679.92
Brunei	BRN	118211.91
United States Virgin Islands	VIR	113333.98
Trinidad and Tobago	TTO	109068.18
Kuwait	KWT	103089.84
Norway	NOR	99923.81

This comparison is useful because it allows countries to be compared on a more equal basis. It shows that differences in total energy use are not only a matter of country size, but also of how much energy each person consumes on average.

6.4 Which countries had the highest electricity carbon intensity and how much did they reduce it?

The final analysis examines countries with the highest electricity carbon intensity at their first available observation and compares that value with their latest available observation. This is useful because it identifies countries that began from a high emissions intensity and whether they later reduced it.

The results show that several countries started with very high carbon intensity values, but not all of them reduced them in the same way. Poland, North Macedonia, and Kiribati stand out as countries with substantial declines, while some smaller territories and countries reported repeated values at 1000.0, suggesting limited variation in the reported series.

Table 5: Countries with the highest initial electricity carbon intensity and their reduction by the latest available year

Country	ISO code	Carbon intensity first year	Carbon intensity last year	Reduction
Turkmenistan	TKM	1306.60 (2000)	1306.27 (2024)	0.33
Uzbekistan	UZB	1103.50 (2000)	1000.00 (2025)	103.50
Saint Helena	SHN	1000.00 (2000)	1000.00 (2023)	0.00
Kiribati	KIR	1000.00 (2000)	500.00 (2024)	500.00
Falkland Islands	FLK	1000.00 (2000)	1000.00 (2023)	0.00
Montserrat	MSR	1000.00 (2000)	1000.00 (2024)	0.00
Poland	POL	966.97 (1990)	588.60 (2025)	378.37
North Macedonia	MKD	925.22 (1990)	441.35 (2025)	483.87
Mongolia	MNG	910.35 (2000)	816.31 (2025)	94.04
Bahrain	BHR	904.76 (2000)	902.24 (2024)	2.52

This finding is important because it shows that some countries have made significant progress in reducing the emissions intensity of their electricity systems, while others remain close to their initial levels. These differences reflect the uneven pace of the energy transition and the continuing role of fossil-based electricity generation in some countries.

7 Conclusion

This project analysed renewable electricity, fossil-fuel dependence, energy consumption per person, and electricity carbon intensity across countries. The results showed that renewable energy shares and energy use vary considerably between countries, while several countries also reduced their fossil-fuel dependence and carbon intensity over time.

The project also demonstrated a complete data workflow: the raw dataset was validated and cleaned with Python and pandas, loaded into PostgreSQL, and analysed using SQL. BigQuery was explored as a cloud-based extension. The results should be interpreted carefully because some countries and years have more complete data than others.

References

- [1] Hannah Ritchie, Pablo Rosado, and Max Roser. Energy. *Our World in Data*, 2023. <https://ourworldindata.org/energy>.